Error Correction

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My proposed encoding

define a system with a conserved or approximately conserved excitation band (or fixed total excitation sector).
Split the system into two spatial regions:

- Left half L
- Right half R

Define:

- N_L = number of excitations in left
- N_R = number of excitations in right

Logical basis:

- $|0_L\rangle$: states with $N_L < N_R$
- $|1_L\rangle$: states with $N_L > N_R$

We are sector confined

Example $b = 5$, $N = 10$

$10\,000\,|\,01111\rangle$ } $|0\rangle_L$

$N_L = 1 < N_R = 4$

$1111\,0\,|\,00001\rangle$ } $|1\rangle_L$

$N_L = 4 > N_R = 1$

Code space:

$\mathcal{C} = \text{span}\{|0_L\rangle, |1_L\rangle\}$

Both lie in:

- same total excitation sector
- same energy band (important)

$N_L = \sum_{i \in L} n_i \qquad N_R = \sum_{i \in R} n_i$

Imbalance $M = N_R - N_L$

Define binary encoding

$|0_L\rangle \in \mathcal{H}_{M \le 0}$

$|1_L\rangle \in \mathcal{H}_{M > 0}$

We define our error set

$\mathcal{E} = \{X_i, Y_i, Z_i\}_{i=1}^{N}$ → complete for Markovian noise (citation needed)

Because logical structure is defined by sign of M, every operator must be understaood in terms of how it changes M

Classical Memory

1. Z_i

$[Z_i, N_L] = 0 = [Z_i, N_R] \Rightarrow [Z_i, M] = 0$

$Z_i\,|10001011117\rangle = |10001011117\rangle$ ($M = 3 > 0$)

$Z_i|0\rangle_L = |0\rangle_L$

$Z_i|1\rangle_L = |1\rangle_L$ } $\forall i$

Z errors are purely dephasing, DFS handles them

2. X_i

$X_i : |0\rangle \to |1\rangle$

$i \in L : N_L \xrightarrow{X_i} N_L \pm 1$

$i \in R : N_R \xrightarrow{X_i} N_R \pm 1$ } $\Rightarrow M = N_R - N_L \to N_R - (N_L \pm 1) \Rightarrow M \to M \pm 1$

After 1 local errors

$M = 3 \longrightarrow$ 2 or 4

After k local errors

$M \longrightarrow 3 \pm k$

$\Rightarrow$ logical flip requires

$k = M$ local X_i errors

Would not work without confinement

n being conserved would not work without sector confinement.

All of this rests on the fact that local X errors do not scramble since n is a quasi-conserved quantity

Also the robustness of this code increases with system size, so it's scalable

3. $Y_i = iX_iZ_i$

For this example Y_i behaves like X_i

Say we wanted to encode

$$|\Psi\rangle = \frac{|0\rangle_L + |1\rangle_L}{\sqrt{2}}$$ → Quantum information not classical

1. Z_i

$Z_i|0\rangle = |0\rangle \quad Z_i|1\rangle = -|1\rangle$

- it gives a **phase on occupied sites only**
- it does not change excitation number
- crucially: it is **local**, not global.

$Z_i|n\rangle = |n\rangle$

$Z_i|\Psi\rangle = |\Psi\rangle$

2. X_i

$X_i|\Psi\rangle = |\Psi\rangle$ if $M > $ local X_i errors

So quantum memory preserved using this encoding